Unit 1: Using Objects and Methods **ANSWER KEY**
Cut-And-Paste Object Creation Activity

Step 1: Cut out the code snippets that were given to you. You will arrange the code pieces to make three object classes: `Elephant`, `Giraffe`, and `Lion`. Arrange them in the correct order and glue them to a new piece of paper.

```
public class Elephant {
  private int age;
  private double weight;

  public Elephant(){
    age = 0;
    weight = 0;
  }

  public Elephant(int a, double w){
    age = a;
    weight = w;
  }

  public void loseWeight(){
    weight -= 10.5 * age;
  }

  public double calculateFoodNeeded(int days) {
     return weight * 0.05 * days;
  }

  public void talkingElephant(){
    System.out.println("The elephant's age is " + age);
    System.out.println("The elephant's weight is " + weight);
  }
}
```

```
public class Giraffe {
  private double height;
  private double neckLength;

  public Giraffe() {
    height = 4.0;
    neckLength = 0.75;
  }

  public Giraffe(double h, double n) {
    height = h;
    neckLength = n;
  }
```

```
  public void growTaller() {
    height += 0.5;
    neckLength += 0.2;
  }

  public double heightAfterYears(int years, double growthRate) {
      return height + (years * growthRate);
  }

  public void displayGiraffe() {
    System.out.println("The giraffe's height is " + height);
    System.out.println("The giraffe's neck length is " + neckLength);
  }
}

public class Lion {
  private boolean gender;
  private double roarVolume;

  public Lion() {
    gender = true;
    roarVolume = 80.0;
  }

  public Lion(boolean g, double v) {
    gender = g;
    roarVolume = v;
  }

  public void roarLouder() {
    roarVolume += 5;
  }

   public double roaringDistance() {
    return roarVolume * 1.5;
   }

  public void showLion() {
    System.out.println("The lion roars at " + roarVolume + " dB.");
  }
}
```

Step 2: After you have created your three classes, complete the driver class below by filling in the blanks with the appropriate code.

```
public class ZooDriver {

  public static void main(String[] args) {

    // Create objects using parameterized constructors
    Elephant dumbo = new Elephant(10, 7000.0);
    Giraffe jeff = new Giraffe(5.0, 1.75);
    Lion simba = new Lion(false, 85.0);

    // Print initial values
    System.out.println("Initial animal stats:");
    dumbo.talkingElephant();
    jeff.displayGiraffe();
    simba.showLion();

    // Call methods to modify attributes
    dumbo.loseWeight();
    jeff.growTaller();
    simba.roarLouder();

    // Print updated values
    System.out.println("\nUpdated animal stats:");
    dumbo.talkingElephant();
    jeff.displayGiraffe();
    simba.showLion();

    // Call special methods
    System.out.print("Your elephant will need ");
    System.out.print(dumbo.calculateFoodNeeded(5));
    System.out.println("lbs of food for 5 days.");

    System.out.print("Your giraffe will be ");
    System.out.print(jeff.heightAfterYears(4, 0.25));
```

```
        System.out.println(" after 4 years and a 25% growth rate.");

        System.out.print("You can hear your lion roar ");
        System.out.print(simba.roaringDistance());
        System.out.println(" meters away.);
    }
}
```

Step 3: Answer the questions below using your three object classes and your driver class.

1) What are the initial values of each object when they are first created?

Note: This will depend on what students use when they create their objects. I let them choose numbers that make sense!

My elephant has an age of 10 and a weight of 7000 lbs. My giraffe is 5 meters tall and has a neck length of 1.75 meters. My lion is male (boolean value of false) and has a roar volume of 85 dB.

2) What is one important piece of information that we are missing in our three classes that can be solved by adding in comments to the code?

Units! We do not know any units (although, we could guess), and we do not know what the boolean gender stands for. Comments can help.

3) After calling the `loseWeight()` method on `dumbo`, what happens to the weight attribute?

Dumbo's weight will decrease by 10.5 times his age. His weight is currently 7000, and his age is 10, so 7000 – 10.5 * 10 = 7000 – 105 = 6989.5

4) What happens to the `height` and `neckLength` of `jeff` after calling `growTaller()`?

Height (which is currently 5m) will increase by 0.5m and become 5.5. Neck Length, which is currently 1.75m, will increase by 0.2m and become 1.95.

5) What is the new `roarVolume` of `simba` after calling `roarLouder()`?

Simba's roarVolume (which is currently 85) will increase by 5 to 90.

6) If we create a new `Giraffe` using `Giraffe babyGiraffe = new Giraffe();`, what will be its values for `height` and `neckLength`?

We will use the default constructor which makes height 4 and neckLength 0.75.

7) What will be displayed when each of the lines of code are run under "call special methods"?

`dumbo.calculateFoodNeeded(5)` will display the result of 6989.5 * 0.05 * 5, which is 1747.375.
`jeff.heightAfterYears(4, 0.25)` will display the result of 5.5 + (4 * 0.25) which is 6.5
`simba.roaringDistance()` will display the result of 90 * 10.5 which is 945.